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REPORT

Shrines in Central Italy conserve plant diversity and large trees

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Abstract Sacred natural sites (SNS) are instances of biocultural landscapes protected for spiritual motives. These sites frequently host important biological values in areas of Asia and Africa, where traditional resource management is still upheld by local communities. In contrast, the biodiversity value of SNS has hardly been quantitatively tested in Western contexts, where customs and traditions have relatively lost importance due to modernization and secularization. To assess whether SNS in Western contexts retain value for biodiversity, we studied plant species composition at 30 SNS in Central Italy and compared them with a paired set of similar but not sacred reference sites. We demonstrate that SNS are important for conserving stands of large trees and habitat heterogeneity across different land-cover types. Further, SNS harbor higher plant species richness and a more valuable plant species pool, and significantly contribute to diversity at the landscape scale. We suggest that these patterns are related not only to pre-existent features, but also to traditional management. Conservation of SNS should take into account these specificities, and their cultural as well as biological values, by supporting the continuation of traditional management practices.

Keywords Biocultural conservation · Biodiversity · Central Italy · Old-growth forests · Sacred natural sites · Traditional management

INTRODUCTION

It is recognized that the fate of biodiversity will increasingly depend on conservation in anthropogenic landscapes, within or outside protected areas (PAs) (Willis et al. 2012). Partly, this is due to limitations of PA networks (Joppa and Pfaff 2010; Guarino et al. 2015). Partly, there is growing appreciation of the role played by cultural landscapes in supporting habitats and species (Palang et al. 2004). In some instances, traditional management is even key to maintaining species-rich grasslands (Maurer et al. 2006) and agro-forestry matrices (Parrotta and Agnoletti 2007).

Sacred natural sites (SNS) are primary examples of biocultural landscapes, where biodiversity is sustained by cultural practices (Verschuuren et al. 2010; Pungetti et al. 2012). The International Union for the Conservation of Nature (IUCN) defines SNS as "areas of land or water having special spiritual significance to peoples and communities" (Wild and McLeod 2008). They are found in every continent except Antarctica (Bhagwat and Rutte 2006) and in relation to both indigenous and mainstream faiths (Dudley et al. 2009). It has been suggested that SNS resemble community-based resource management and common-pool resources (Berkes 1999; Rutte 2011), where conservation or sound use of biodiversity is embedded in traditional institutions and beliefs (Colding and Folke 2009). A growing literature has demonstrated that in parts of East Asia and Africa SNS host higher biodiversity than surrounding areas and even PAs (reviewed in Dudley et al. 2010). A similar link, in contrast, has hardly been quantitatively tested in European contexts, although having been addressed in case studies (Mallarach and Papayannis 2010; Frascaroli 2013). Given the dependence of much of biodiversity in Europe on traditional knowledge and management (Otero et al. 2013), it is surprising that the potential of European SNS as "biocultural refugia" (Barthel et al. 2013) has not been more thoroughly investigated.

In a preliminary study (Frascaroli 2013), it was shown that an association with natural areas is very common for Catholic sites in Central Italy. This region hosts

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outstanding religious heritage and is indicated as one of the main biodiversity hotspots in Europe, largely in virtue of high plant diversity and endemism (Myers et al. 2000). Here, we quantitatively assess the biodiversity value of SNS in the same area, as compared to analogous non-sacred reference sites. We hypothesize that SNS have higher plant species diversity than reference sites in virtue of having been protected for spiritual reasons and that SNS with more religious importance have higher diversity than the less important ones. This last hypothesis is based on functional theories in anthropology (Rappaport 1999), which view religion as reinforcing cooperative behavior, leading to sounder management of communal resources (Sosis and Ruffle 2003; Gómez-Baggethun et al. 2012). SNS with more religious importance may thus be expected to have been better protected from disruptive human uses or overharvesting, promoting the persistence of species vulnerable to disturbance or with reduced dispersal abilities. The specific objectives of this paper are:

(1) To establish whether SNS perform better than reference sites as to habitat heterogeneity, plant diversity, and conservation of large trees (i.e., ≥40 cm diameter at breast height, DBH);
(2) To determine how SNS and reference sites contribute to both local and overall species pools, with particular emphasis on endemic and Mediterranean species, and species representative of conservation priority habitats;
(3) To assess the background mechanisms driving the possible differences across SNS and reference sites.

MATERIALS AND METHODS

Study area and sites

Central Italy consists of the regions Tuscany, Marche, Umbria, Lazio, Abruzzo, and Molise (Fig. 1). Geomorphology is dominated by hills (62.4 %) and mountains (34.2 %), whereas plains are few (3.3 %) and limited to the coastline and valley bottoms. Around 23 % of the land surface is part of some conservation scheme: ca. 12 % is included in both the Official List of Protected Areas and Natura 2000 network (EC 1992), while ca. 11 % falls within the Natura 2000 network alone.

SNS in the region are characterized by high diversity of function, make-up, and size. They can range from a small chapel or even a simple icon hung on a tree, to large temples surrounded by forest. Some SNS, often founded in the Middle Ages, serve as dwellings for contemplative monks and hermits, and are permanently inhabited by religious communities. Others became worship places (shrines) following the passage of a saint or a miraculous apparition. In general, shrines are not permanently inhabited but visited only on special occasions by believers who come to venerate a holy object, often a natural feature, kept therein. Combinations of the two categories (i.e., shrines tended by monastic communities) are also common. Some SNS were abandoned throughout the centuries due to different reasons but may remain important as historical monuments and landmarks (Frascaroli 2013).

Human artifacts are found at nearly all SNS, whether in the form of buildings or simple structures carved into the rock. In a very few cases, SNS consist only of sanctified natural features (Fig. 1). In general, however, the natural areas around the artifacts are also considered sacred and protected, and can thus host important biodiversity (Dudley et al. 2009). The size of these areas varies considerably, ranging from a fraction of a hectare for the smaller shrines to a few hundreds hectares for the larger monastic estates (Frascaroli 2013). In most cases, however, it is not possible to determine the areal extent of SNS with reasonable confidence, due to two reasons: (1) the lack of clear borders delimiting their natural areas; (2) SNS in Central Italy, differently from elsewhere (e.g., Aerts et al. 2006), are frequently included in broader areas with similar land-cover. This may be a consequence of relatively recent processes of reforestation that have reconnected previously isolated forest fragments following the abandonment of rural areas (Amici et al. 2015).

In this study, we focused exclusively on shrines because they are most similar to the model of community-based resource management that characterizes SNS in other parts of the world (Rutte 2011). Shrines are often associated with popular devotions and rural livelihoods. They retain considerable importance for local communities, although with varying degrees of devotion. Celebrations and pilgrimages on foot are still carried out at most shrines at least once per year, while tourists without religious motivations are a minority of all visitors. Also, it is not uncommon that shrines are directly tended by local people rather than religious or bureaucratic institutions (Frascaroli 2013). Possible effects on conservation, therefore, are firstly the product of traditional management informed by local customs and traditions. In contrast, PAs privilege species management in forest areas, with special reserves managed with minimal intervention (Götmark 2013). Promoting economic development via tourism in the more inhabited areas is also a common objective of parks and reserves in the region.

We identified research sites relying on both a preliminary study (Frascaroli 2013) and the database of Christian shrines in Italy (CSC 2003). We considered as suitable study sites all shrines located in natural settings, such as forests, cliffs, and mountain grasslands. We identified a sample of 30 SNS responding to these criteria: (1) even

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Fig. 1 Map of Italy showing the five regions that compose the study area and the 30 sacred natural sites in the sample. Each site is marked by a unique ID, with site details provided in Table S1. Examples of four sacred sites are visible in the pictures accompanying the map, clockwise from *top-left*: **a** Leccio delle Ripe (St. Francis' Holm-oak), which also represents the only sample site without built structures; **b** hermitage of St. Mary of Acquarella; **c** Water St. Franco; and **d** shrine of the Very Holy Trinity of Vallepietra

distribution across the five administrative regions; (2) comparable number of SNS within and outside PAs; (3) representation of the different vegetation types found at SNS in the area; and (4) representation of a continuum of religious importance, assessed on the basis of interviews with shrine custodians and ordered in four categories from

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1 (nearly abandoned SNS) to 4 (hubs attracting ten or more thousand pilgrims a year). We thus included 20 SNS located in different types of forest, 3 grassland SNS, and 7 SNS located in open-canopy forests or mixed forest-grassland mosaics (see Table S1 for site details). These proportions approximate the overall distribution across vegetation types of SNS in Central Italy (Frascaroli 2013).

Finally, we paired each sample SNS with a comparable reference site in a non-sacred area nearby (mean distance ± SD: 534 ± 323 m), having similar elevation (mean elevation difference: 55 ± 45 m), aspect, and vegetation type (e.g., *Quercus ilex* forest). We used geographic information system (GIS) software (ESRI 2010), satellite-based land-cover maps (ESA 2010), and a geographic position system (GPS) device to identify suitable reference sites for each SNS. When several options were available, we randomly picked one after discarding those that could not be accessed by road or walking trails. In six instances, we found that the exact vegetation type at the SNS was the only occurrence in the landscape. In those cases, we had to select reference sites that matched SNS as closely as possible, but differed slightly as dominant vegetation (Table S1). Nonetheless, replicating the analyses without these pairs of sites did not significantly alter the results, confirming that our conclusions are robust to these minimal deviations.

Table 1 Classification of local scale land-cover types, and number of occurrences of each type at sacred natural sites and reference sites

Type of land-cover	Number of occurrences at sacred sites	Number of occurrences at reference sites
Anthropogenic		
Barren/degraded	4	0
Dirt road/path	22	12
Paved road/car park/built structure	9	4
Total occurrences	35	16
Natural		
Broadleaf closed-canopy forest	26	22
Broadleaf open-canopy forest	2	5
Cliffs/rock outcrops	22	11
Dry grassland	7	7
Grotto	1	0
Mediterranean scrub	11	12
Mixed broad- and needle-leaf forest	4	4
Mountain pasture	2	2
Mountain scrub	3	2
Orchard/olive grove/lawn	5	1
Riparian zone	3	2
Spring/water course	6	1
Wet meadow	1	0
Total occurrences	93	69

Vegetation sampling and data collection

Given the difficulty to determine the borders of SNS in the region, we considered a 25 m buffer around the perimeter of each shrine as area of influence, leading to variable areas according to the size of the shrine. This was likely an underestimation of the real extent of most SNS, but provided a safe standard for comparing sites across the study system.

To sample trees in those areas, we laid one to three 25 m × 4 m transects at each SNS. We recorded the species and measured DBH of all mature trees (≥10 cm DBH) rooted within. The number of transects at each site varied according to local geomorphology: while aiming to maximize sampling intensity, our efforts were often limited by natural obstacles (e.g., the presence of cliffs). The transects were laid adjacent to each shrine where the natural patch started, and stretched 25 m away from it. When feasible, orientation of the first transect was randomly determined, and the following ones positioned so as to evenly divide the area.

To sample understory vegetation, three 1 m × 1 m plots were located at each end and in the middle of the transects, for a total of 3–9 1 m^2 plots at each SNS. Species identity and estimated cover were recorded for all vascular plants inside the plots, including herbaceous and shrub layers and tree canopy projections. Plant specimens were collected and dried for subsequent identification, which was based on Pignatti (forthcoming).

As an indication of local habitat composition, we recorded the occurrence of different land-cover types within the 25 m buffer and estimated their percentage cover. Sixteen categories were used for this classification (Table 1).

The same design used at each SNS, including extent of sampled area, number, and collocation of plots, and assessment of habitat composition, was replicated at the paired reference sites. Overall, 63 transects and 189 plots were laid across the SNS (mean ± SD: 2.1 ± 0.7 transects per site), and as many across the reference sites. In order to identify most of the plant species, data collection was conducted in the period late May–August, with each pair of sites sampled in the same week and sites at lower elevations sampled earlier in the season. Fieldwork was carried out in summers 2011–2012.

Statistical analyses

Number of land-cover types, natural habitat heterogeneity (Shannon–Weiner's H'), and proportion of anthropogenic area were calculated from the habitat assessment data at

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each site. To quantify heterogeneity at the landscape scale, instead, we used GIS. We recorded the number of different *GlobCover 2009* (ESA 2010) land-cover types in a radius of 1000 m around each site and calculated Shannon–Weiner's H' based on the area of each land-cover. Tree density (stems ha^{-1}), basal area ($m^2\ ha^{-1}$), and species richness were calculated from the transect data at each site. Plant species richness per site, plant species richness per plot (1 m^2), and within-site β-diversity (1 − Jaccard Index of similarity between all plots at a site; Magurran 1988) were calculated from the 1 m^2 plot data.

To compare metrics between SNS and reference sites, we used paired t tests whenever the assumptions of normality and equal variance of the data were respected. Otherwise, we used Wilcoxon sign-rank tests for two related samples or generalized linear models.

To assess the influence of other variables on plant richness at all sites, we performed analysis of variance (ANOVA) on a linear model. To respect the assumptions of normality and equal variance, we used a square root transformation of the response variable, based on the Box–Cox transformation technique. The environmental variables in our model included habitat and landscape heterogeneity, altitude, tree density, tree basal area, and dominant vegetation type (factor with six levels). Anthropogenic variables included religious importance and distance from the closest PA (previously measured in GIS). As a last source of variability, we considered site location as described by two factors: administrative region (five levels) and a unique locality name for each pair of sites (30 levels). We did not include site area as a variable in our analysis because of the problems associated with area delimitation (as specified above). Its effects, however, were accounted for through design, by sampling equal areal extents at each pair of SNS and reference site, and by choosing pairs of SNS and reference sites that were part of the same broader natural area. Further, when equalizing sampling intensities across sites in our model, we avoided using rarefaction by area, as this technique is not robust to variations in sampled area (Chiarucci et al. 2009). Rather, we considered as sampling unit the pooled richness of the three plots within each transect and calculated the mean pooled richness for all transects (1–3) at each site.

Finally, we divided all the recorded species according to life form, phytosociological preference, and biogeographical distributional range (chorology), based on the available literature (Ellenberg 1996; Aeschimann et al. 2004; Guarino et al. forthcoming). The phytosociological classification of habitats is coherent with the conservation targets of the European Community's Natura 2000, so it was used to determine what taxa are representative of priority habitats (Blasi et al. 2010). We used this information to analyze differences in community composition and the distribution of plant groups of conservation interest (i.e., plants with an endemic and Mediterranean distribution, or representative of Natura 2000 priority habitats) across SNS and reference sites. To understand the relation between environmental gradients and composition of plant communities, and assess whether the environmental variables previously used as predictors of species richness also explain composition, we performed Detrended Correspondence Analysis (DCA) on a square root transformation of the plant cover data within the plots.

When multiple pairwise comparisons were performed on the same sample data, we applied Bonferroni correction to the resulting p values. All statistical analyses were performed with the software R version 3.1.0 (R Core Team 2014).

RESULTS

Comparison between sacred and reference sites

Landscape and habitat heterogeneity

At the landscape scale, we found only negligible differences in land-cover heterogeneity around SNS and reference sites (mean ± SE: 0.78 ± 0.08 vs. 0.86 ± 0.09, $p = 0.327$). At the local scale, in contrast, we recorded a larger number of land-cover patches at SNS than reference sites (Table 1), and both heterogeneity and number of natural land-cover types per site were noticeably higher at SNS (0.74 ± 0.05 vs. 0.45 ± 0.06, $p < 0.001$, and 3.1 ± 0.16 vs. 2.3 ± 0.17, $p < 0.001$, respectively). The proportion of anthropogenic area was also higher at SNS (10.2 ± 2 vs. 2.3 ± 0.7 %, $p < 0.001$), although varying in relation to religious importance (Fig. S1). This proportion was nearly 31 % (±6.6 %) at the SNS with highest religious importance, while ranged between 3.6 % (±2.3 %) and 8.1 % (±2 %) at the less important ones. Neither evident correlation was found between the number of land-cover types at the local and landscape scale, nor between local- and landscape scale heterogeneity ($R^2 = 0.006$, $p = 0.546$, and $R^2 = 0.0009$, $p = 0.822$, respectively), indicating that different processes drive habitat and landscape structure at fine and coarse scale.

Tree size and forest structure

While similar mean tree density (ca. 850 stems ha^{-1}) was recorded at both SNS and reference sites, large trees occurred more frequently at SNS (mean ± SE: 78 ± 18 vs. 28 ± 8, $p < 0.001$, Fig. 2). Tree biomass as estimated by basal area was also considerably larger at SNS (54 ± 7.3 vs. 27 ± 3.3 $m^2\ ha^{-1}$, $p < 0.001$). In contrast, only a marginally higher tree species richness was recorded at SNS than reference sites (2.5 ± 0.3 vs. 2.1 ± 0.2, $p = 0.152$).

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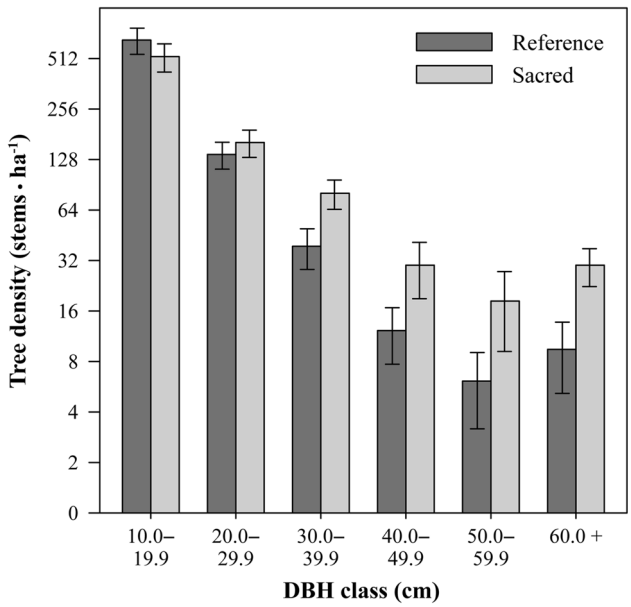

Fig. 2 Mean density (stems ha^{-1}) of trees in different diameter classes at sacred and reference sites (two-way ANOVA of quasi-Poisson GLM testing distribution across sacred and reference sites of trees <40 cm DBH: $F = 0.464$, $p = 1$; two-way ANOVA of quasi-Poisson GLM for trees ≥40 cm DBH: $F = 20.412$, $p < 0.001$). *Error bars* indicate standard error of the mean

Plant diversity

Compared with reference sites, SNS hosted significantly higher plant species richness both per site (mean ± SE: 19.6 ± 1.7 vs. 15.7 ± 1.5, $p < 0.01$) and per plot (9.9 ± 0.9 vs. 8 ± 0.8; $p < 0.05$). Higher β-diversity was also recorded within SNS (0.73 ± 0.02 vs. 0.65 ± 0.03, $p < 0.01$), indicating greater variation in species composition between the plots at each site.

Factors related to plant diversity

Plant species richness per sampling unit varied significantly across geographic regions and locations, as well as vegetation types (Table 2). Greatest richness was found in grassland sites, followed by open-canopy *Q. ilex*-dominated forests (mean ± SE: 17.9 ± 1.8 and 14.8 ± 1.9, respectively; Fig. 3). *Fagus sylvatica* and closed-canopy *Q. ilex* forests, in contrast, were the least speciose habitats (7.6 ± 1.4 and 7.9 ± 0.7, respectively). Species richness of other deciduous forest assemblages was similar whether they had open or closed canopy (12.5 ± 1.7 and 12.6 ± 0.9), although with different variability. SNS had higher or slightly higher mean richness than reference sites within each vegetation type.

Altitude did not significantly affect plant species richness, while there was a significant effect of habitat heterogeneity at both local and landscape scales, as more heterogeneous habitats and landscapes supported more speciose plant assemblages. Tree density and basal area, in contrast, negatively affected plant richness (Fig. S2).

As mentioned, SNS had significantly higher mean species richness than reference sites, but the magnitude of this difference varied according to SNS' religious importance, following a bell-shaped trend (Fig. 4). The difference was negligible or minor for abandoned and very important SNS, while it peaked for important or moderately important SNS. The higher mean species richness of abandoned SNS as compared to other SNS, instead, seemed mostly a

Table 2 Summary table of sequential sum of squares ANOVA, testing the incidence of anthropogenic variables, environmental variables, and geographical location on plant species richness per sampling unit (3 m^2)

Explanatory variable	df	SS	*F*	*p*
Site type (sacred vs. reference)	1	1.02	10.93	0.0045*
Religious importance	1	1.41	5.02	0.0122*
PA	1	0.07	0.76	0.3970
Distance from PA	1	0.18	1.88	0.1893
H′ natural habitats	1	1.90	20.28	0.0004*
H′ landscape	1	0.51	5.48	0.0326*
Altitude	1	0.00	0.00	0.9918
Tree density	1	6.21	66.35	<0.0001*
Tree basal area	1	0.67	7.18	0.0164*
Vegetation type	5	5.47	11.69	<0.0001*
Region	4	4.65	12.41	<0.0001*
Location	23	11.55	5.61	0.0005*
Site type: distance form PA	1	1.08	11.54	0.0037*
Residuals	17	1.50		

* Significant variables ($p \leq 0.05$)

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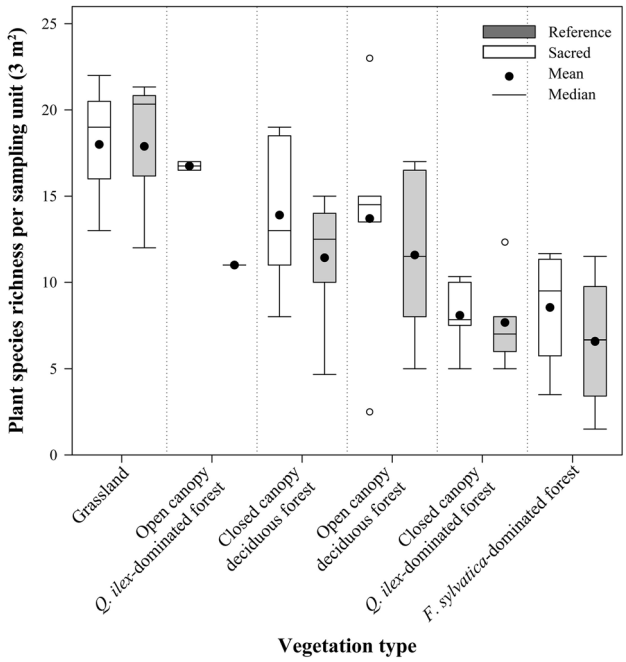

Fig. 3 *Boxplots* of plant species richness per sampling unit (3 m^2) across vegetation types at sacred and reference sites, summarizing median, mean, upper and lower quartiles, and minimum and maximum data values

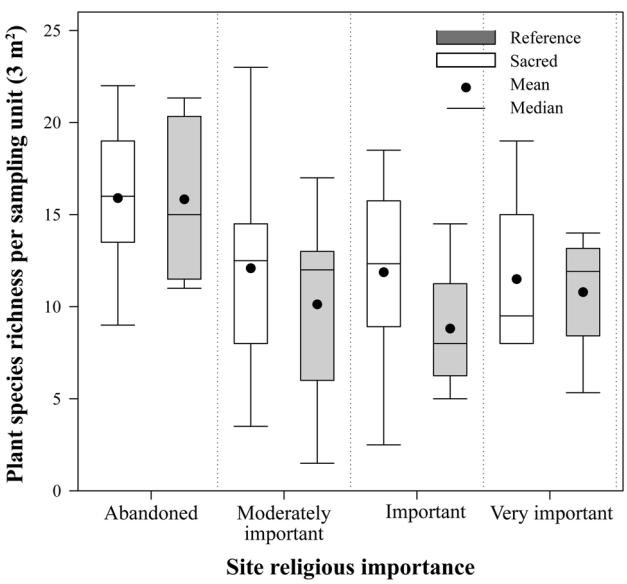

Fig. 4 *Boxplots* of plant species richness per sampling unit (3 m^2) across categories of religious importance at paired sacred and reference sites, summarizing median, mean, upper and lower quartiles, and minimum and maximum data values

random effect of site selection, as 40 % of abandoned SNS were located in species-rich grasslands.

The influence of PAs was also uneven. We found nearly no difference between sites located within and outside PAs (mean ± SE: 11.7 ± 0.8 and 11.7 ± 1.1, respectively), but

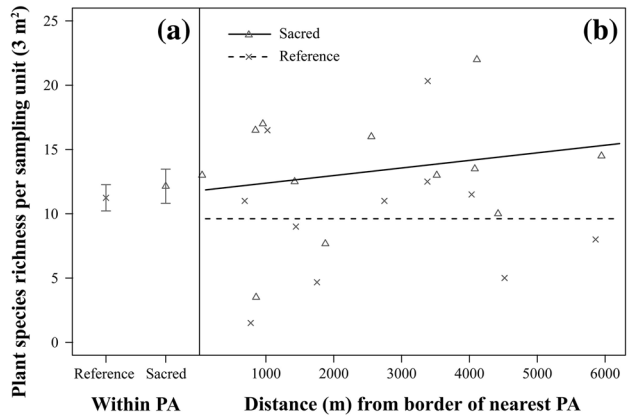

Fig. 5 Mean plant species richness per sampling unit (3 m^2) at sacred and reference sites within protected areas (PAs) (**a**), and relationship between plant species richness per sampling unit and distance from PA at sacred and reference sites outside PAs (**b**). *Error bars* indicate standard error of the mean

there was an interaction between official protection and protection based on "sacredness." The difference in plant richness between SNS and reference sites was negligible within PAs (12.1 ± 1.3 and 11.2 ± 1, respectively), but progressively widened at increasing distance from PAs (Fig. 5).

Distribution of plant species and plant species composition across sacred and reference sites

We recorded 352 different plant species in the 378 study plots. Of these species, 128 were found uniquely at SNS, 79 were limited to reference sites, and the remaining 145 were shared by both. Considering pairs of SNS and reference sites as units, we recorded a mean richness of 28.1 (SE: ±2.5) species per pair. The contribution to this figure was different for types of site ($p < 0.001$), as nearly half (mean ± SE: 44.1 ± 2.2 %) of the species at each pair were unique to SNS, 29.5 % (±2.7 %) were unique to reference sites, while the remaining 26.4 % (±2.2 %) were shared. This was not just a byproduct of higher species richness at SNS: also the proportion of unique species out of species richness at each site was significantly higher for SNS than reference sites (62.9 ± 2.5 vs. 52.3 ± 3.7 %, $p < 0.05$).

At least 28 % of the 128 species unique to SNS were representative plants of Natura 2000 priority habitats (Table S2). Furthermore, plants in two priority groups (Asplenietea trichomanis and Adiantetea) were found exclusively at SNS. On average, the number of species representative of Natura 2000 priority habitats was marginally higher at SNS than reference sites (mean ± SE: 10.3 ± 0.7 vs. 8.9 ± 0.7). Similarly, of 91 species having a Mediterranean distributional range, 40 % were found exclusively at SNS (against 17 % at reference sites),

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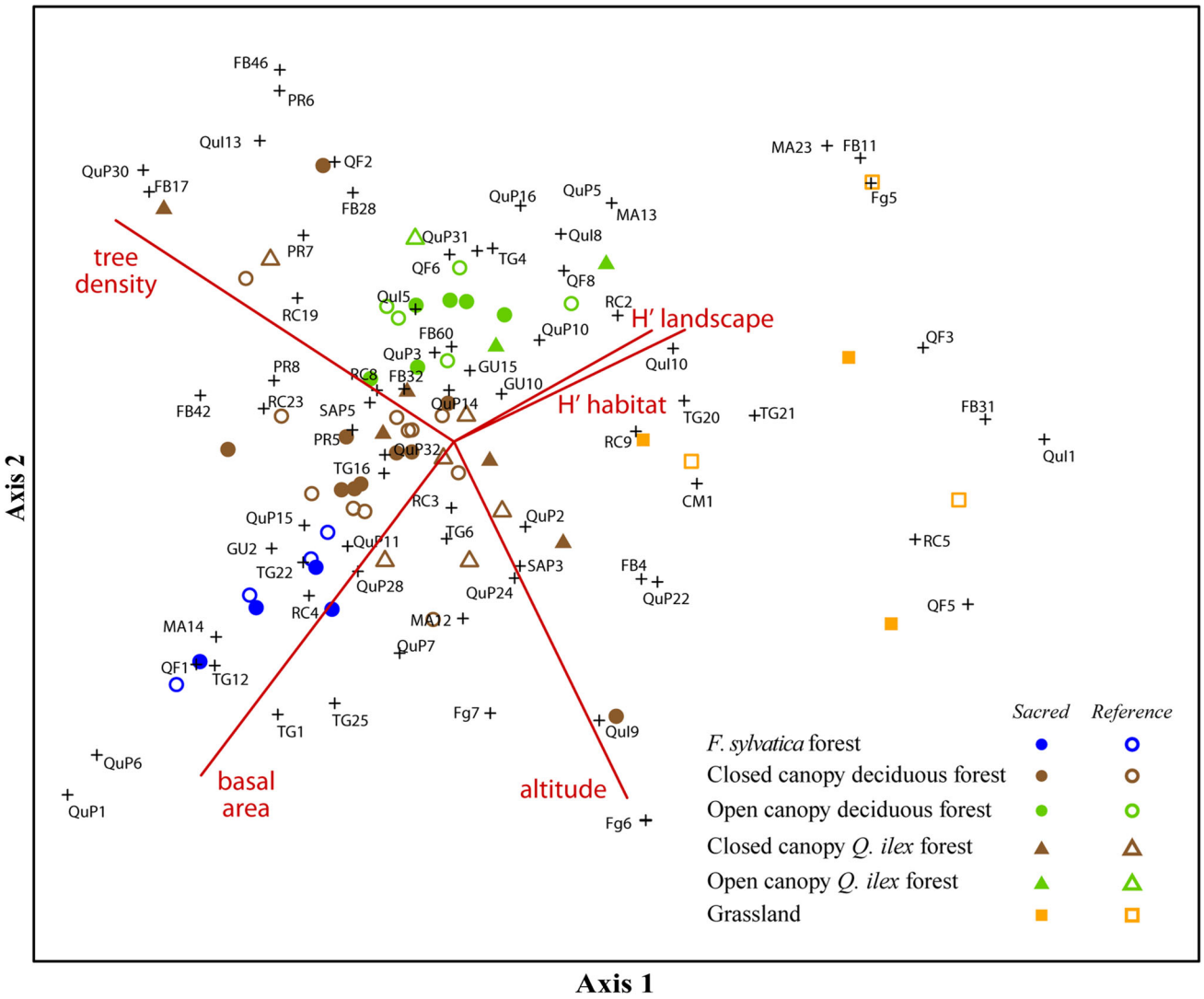

Fig. 6 Detrended Correspondence Analysis displaying the study sites and most abundant species. The analysis shows a gradient between forest and grassland habitats along Axis 1 (*left to right*) and a distinction between different types of forest along Axis 2. Plant species are labeled with alphanumeric IDs that also indicate phytosociological group. Full species names and characterization are provided in Table S2

including 5 endemic taxa (against 2 found uniquely at reference sites and 3 shared by both). Finally, SNS hosted a larger number of Eurasiatic species (56 vs. 35), while there was nearly no difference in the distribution of European and cosmopolitan plant species.

The DCA indicated that the environmental variables used as predictors of species richness are also related to changes in species composition (Fig. 6). Axis 1 captured a gradient between plants associated with forest habitats at low axis values (e.g., *Fraxinus ornus* and *Quercus cerris*), and those associated with non-forest habitats at high values (e.g., *Bromus ramosus*, *Cistus incanus*, and *Silene vulgaris*). Axis 2, instead, captured a gradient internal to forest habitats. High axis values correspond to evergreen forest/maquis communities with high stem density (e.g., *Pistacia terebinthus* and *Quercus pubescens*), while deciduous forest communities with fewer but larger trees and located at higher altitudes are represented by low axis values (e.g., *Carpinus betulus* and *Ostrya carpinifolia*).

Although compositional structure was similar for SNS and reference sites as a consequence of study design, some important differences were recorded. Herbs, and especially perennial ones, accounted for a large proportion of plant richness at SNS (mean ± SE: 45 ± 4 and 40 ± 4 %, respectively), whereas herbs, trees, and shrubs contributed nearly equally to plant richness at reference sites (33 ± 4, 35 ± 4, and 32 ± 3 %, respectively). Also, the mean cover of Galio-Urticetea species in the 1 m^2 plots was nearly null

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at reference sites (0.26 ± 0.14 %) while amounted to 4.1 % (±1.4 %) at SNS ($p < 0.001$). Finally, shrub and woody species had very similar cover between SNS and reference sites when all plots were considered (10 ± 2.4 vs. 10.1 ± 2.1 %), while their cover was significantly higher at reference sites if only plots dominated by deciduous forest (14 ± 3.8 vs. 5.5 ± 2.5 %, $p < 0.05$) or grassland (5.6 ± 2.5 vs. 1.5 ± 0.6 %, $p < 0.05$) were considered.

DISCUSSION

SNS are culturally relevant areas that have been used for religious purposes, often for many centuries. Around the world, these sites are frequently associated with high biodiversity and traditional ecological management (Verschuuren et al. 2010; Pungetti et al. 2012). Our analyses yielded similar results for a relatively modernized Western context, suggesting that as socio-ecological systems SNS have been largely resilient to general secularization and modernization.

SNS in our sample are currently part of much vaster continuous areas, exemplified by our reference sites, having similar land-cover or even vegetation type. As such, SNS are not "natural" islands in otherwise degraded areas, although in several instances we found that they have conserved vegetation types that do not occur in the wider landscape. Based on design, our study did not quantify these contributions to landscape scale conservation of habitats. It rather captured fine-grained variations between SNS and surrounding areas, demonstrating that SNS are qualitatively different patches of the same habitat type (cf. Salick et al. 2007), and contribute to local variability, that is, one of the major drivers of regional plant diversity (Valdés et al. 2015). Indeed SNS harbor higher habitat heterogeneity, species richness, and number of large trees. They also contribute larger proportions of unique species to landscape scale diversity and, as a consequence of greater species richness, host more valuable species pools. This is indicated by the higher number of plants with endemic and Mediterranean distribution, or representative of Natura 2000 priority habitats (Table S2).

The biological specificity of SNS is likely the outcome of an interplay between original geomorphology and management. On the one hand, SNS are characterized by abiotic features, such as cliffs, water sources, and grottos, that are absent or sporadic at reference sites. This supports the observation that sacred sites are often established on pre-existing natural landmarks (Frascaroli 2013). The presence of cliffs in particular can explain why plant groups linked to rocky habitats (Asplenietea trichomanis, Adiantetea) or favoring shallow soils (Mediterranean taxa in general) were found exclusively or prevailingly at SNS. On the other hand, there are strong indications that the biological composition of SNS has been influenced over time by management. This is evident with regards to forest structure. The density of trees ≥40 cm DBH measured at SNS is in line with international definitions of "old-growth" forest (Nilsson et al. 2002) and suggests that ancient forest patches have been conserved at SNS as a result of prohibitions on timber extraction. Similar prohibitions are frequent at SNS around the world and can be motivated by both utilitarian reasons (Tengö et al. 2007) and intangible values (Blicharska and Mikusinski 2014).

Greater heterogeneity at the local scale is another distinctive features of SNS as compared to reference sites, and one of the factors driving their higher species richness. Habitat heterogeneity and species richness at SNS may be effects of later successional stages. Land-cover at SNS has presumably been constant for centuries, while our tree data suggest that some reference sites may be the products of relatively recent dynamics of reforestation. However, we found evidence that greater diversity at SNS also depends on continued management and human activities. This is clearly confirmed by the near lack of difference in species richness between reference sites and abandoned SNS, whereas that difference increases for active SNS (Fig. 4).

More in detail, herbivore grazing could explain SNS' greater abundance in Galio-Urticetea species. These are nitrophilous herbs that grow on rich soils, like those forming after prolonged fertilization from animals. Grazing herbivores within forests was common in the Apennines until recently (Manzi 2012), and a direct association between pastoralism and SNS in Central Italy has been documented (Frascaroli et al. 2014). Wood from downed old trees can also concur to form nitrogen-rich soils, but this interpretation seems less likely, clashing with the observation that dead wood tends to be removed from SNS. The lower shrub cover recorded in both deciduous forest and grassland plots at SNS also supports the idea of animal trampling, although the periodic passage of pilgrims and religious ceremonies around the shrines could offer alternative or complementary explanations. Finally, animal husbandry is compatible with signs of active tree management, which we documented at various SNS. Instances of thinning to favor the growth of individual trees, shaping, and pollarding are visible at a number of sites (10) and might be present at others but hard to detect due to discontinued management (Stara et al. 2015). Some of these techniques were widely applied in woodland pastures across Europe, both as a source of fodder (pollarding) and shade (shaping) (Rackham 2006; Agnoletti 2014; Stara et al. 2015). As they tend to positively affect understory growth via increased light availability, they also contribute

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to explaining the greater richness of herbaceous species in forested SNS.

Overall, our results support a view of SNS as biocultural hotspots, in which traditional management contributes to driving habitat and plant diversity. This also provides a possible explanation to the unexpected finding that PA status negatively affects species richness at SNS (Fig. 5). PA management based on minimal intervention (Götmark 2013) may exclude local users from management (Gómez-Baggethun et al. 2010), narrowing the differences between SNS and reference sites within PAs. These differences, in contrast, become wider outside PAs, as traditional management is implemented at SNS but not reference sites. Our data, however, clearly show that these benefits are most distinct within a certain threshold of human activity (Fig. 4). Consistently with our initial hypotheses, important or moderately important SNS favor species diversity more than abandoned ones and are accompanied by only limited negative impacts, as measured by artificial area cover. In contrast, very important SNS confer only marginal benefits with regards to species richness and are surrounded by large artificial areas, like car parks and other services, designed to cater to many visitors. These effects are ecologically expectable, but contradict the hypothesis of a linear relationship between religious importance and conservation benefits. Interestingly, the SNS with highest religious importance are located within PAs (Table S1). They also are the only SNS in our sample to attract significant numbers of tourists, although the majority of visitors remain motivated by spiritual reasons. SNS within PAs in Europe are often promoted as cultural highlights, with consequent increments in visitor numbers (Mallarach and Papayannis 2010). This might contribute to seasonal overcrowding also at the most celebrated of our sites.

Qualitative observations, finally, offer a complementary insight into the relation between religious importance and conservation. Indeed we noted that the SNS with highest religious importance are located in particularly large and valuable natural areas. These are often landscapes where a dominant forest cover (*F. sylvatica* or *Q. ilex*) is intersected by a network of grassland patches. The imprint of pastoralism is evident and celebrated in the symbolism of these SNS (Frascaroli and Verschuuren forthcoming), and animal husbandry still practiced here more than elsewhere in Central Italy. Following Rutte (2011), it can thus be hypothesized that these SNS may have played a prominent role in the management of large silvo-pastoral commons. As such, their contribution to conserving biodiversity should be assessed at a landscape scale, as their actual benefits are confounded by excessive human pressures and not measurable at the local scale.

CONCLUSIONS AND RECOMMENDATIONS

Our results fill an evident gap in knowledge, being among the first to quantitatively prove that SNS host significant biological diversity also in Western non-traditional contexts. SNS, however, are not conservation areas, but rather biocultural hotspots providing a number of ecosystem services that are often crucial to local livelihoods (Dudley et al. 2010). It has been shown that also in Central Italy SNS harbor important ethnobotanical resources, especially related to animal husbandry (Frascaroli et al. 2014). Our study further suggests that SNS in the area might have been important as sources of shade and fodder in summertime woodland pastures. While similar uses may have lost present-day relevance, the intangible benefits (or cultural services) of these SNS remain widely appreciated. According to our observations, they frequently include religious uses, sense of identity, transmission of local knowledge, contribution to social cohesion, and esthetic appreciation. Further, three of the major SNS in our study host water springs that supply the respective watersheds. These multiple values, as well as the views of local stakeholders, should all be considered in management decisions about these sites (Frascaroli and Verschuuren forthcoming).

Limiting the focus to biological management, our data suggest that discontinuing traditional management (Götmark 2013) can negatively affect SNS. Habitat and vegetation diversity at these sites are indeed the outcome of both abiotic factors and active traditional management. Forbidding traditional management to privilege minimal intervention, as commonly done within PAs in the region, will result in assimilating SNS to surrounding areas, with consequent loss of biological distinctiveness and decrease of landscape scale diversity. We rather recommend traditional management to be encouraged at SNS both within and outside PAs, especially when this is supported by local people and traditional uses of the SNS are still alive. Although this option might be hindered by lacking policy instruments (Parrotta and Agnoletti 2007), it would be the most effective way to conserve both nature and culture at some of Europe's last hotspots of biocultural diversity.

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